

HTE staff checklist - Setup

Edition 2026-10-09 | Local OT-2 | 66 first-round + 10 second-round wells

Campaign folder: _____ **Operator/date:** _____

Before use: HTE must release the block/seal/mount, pipetting heights, extraction/IS recovery, LC method/matrix and a measured plan within 12 h. These remain pending; live export enforces LC settings and timing. Use released files.

Before the clock

- Have the real batch/stock/deck CSVs, LC sequences and Python protocols. Check **65 pairs + one L17 reference; all 31 ligands covered in pairs**. Start the local export hook/watcher; key file: `anthropic.key` on the laptop.
- Prepare stocks from `stock_preparation.csv`, including reserves and assay corrections. Prepare ligands in the glovebox. Label IDs/addresses; dispense only homogeneous stocks; cap reserves.

Stock	Preparation / dispensing concentration
SM + naphthalene	250 mM SM + 2.5 mM IS in DCM; prepare at least 5.00 mL.
All 31 ligands	30 mM in toluene; CSV gives each mass and reserve.
Pd(COD)(DQ)	60 mM in toluene; verify identity/assay and solubility.
Toluene	Neat.
Workup / LC diluent	Separate sources: 1:1 water/ACN + 1% v/v HCOOH; no IS.

SM: pyridine-2-sulfonyl fluoride (PyFluor). **Product:** 2-fluoropyridine. Combined-stock recipe at 100% assay: 201.45 mg SM + 250 uL of 50 mM naphthalene/DCM master; DCM to final 5.00 mL in a capped 20 mL glass/PTFE vial. Naphthalene master: 12.817 mg to final 2.00 mL; prepare separate DCM and LC-compatible masters.

OT-2 deck - load stage quantities from `deck_loads.csv`

Left: p20 single GEN2. **Right:** p300 eight-channel GEN2. Manual block moves only.

Slot	Contents
1 / 4	Ligand racks: L01-L24 / L25-L31, at the exact CSV well addresses.
2	12-channel reservoir: A1 SM/IS/DCM; A2 Pd; A3 toluene; A4 workup; A5 LC diluent.
3 / 10	Reaction glass-vial block for dosing/cooling / heater-shaker with released adapter.
6	Second-round substrate array during first dosing; LC plate during workup.
5 / 8 / 9	p20 tips / p300 tips / p300 tips. Keep slots 7 and 11 empty.

LC calibration - prepare stocks now; run standards during round one

Authentic SM/product stocks: **100 mM, 2.00 mL each, no IS** (SM 32.232 mg; product 19.418 mg at 100% assay; correct from COA). Working analyte: 100 uL of 100 mM to 1.00 mL (10 mM). Analytical IS: 40 uL of LC-compatible 50 mM master to 2.00 mL (1 mM).

For **each analyte**, make 1.00 mL per concentration column, with **50 uL of 1 mM IS in every tube**. Use the approved matrix recipe, accounting for stock solvents.

Analyte, uM	0 blank	100	500	1,000	2,500	6,000	5,000 check
Stock, mM	None	10	10	10	100	100	100
Analyte, uL	0	10	50	100	25	60	50
Matrix solvent, uL	950	940	900	850	925	890	900

Fit the blank and five nonzero levels; keep the independent 5,000 uM check outside the fit. **14 injections total**, during the first reaction.

HTE staff checklist - Run

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Block/seal: _____ **LC method/channel:** _____

Follow the local App's pauses and the released CSV well maps. Keep plate orientation and sample IDs intact. Start the competition timer with first-round dosing; stocks and first-batch selection are prepared beforehand.

Step	Staff action
1. First dosing	Import <code>round1/dose_and_react.py</code> into the local OT-2 App. Confirm offsets/deck. Dose 20 uL combined SM/IS/DCM into 66 wells in slot 3.
2. Remove DCM	Manually move the open block to slot 10. Use the released dry-down procedure (initial proposal: 37 C, 300 rpm, 10 min; requires HTE validation). Verify the dry endpoint; return to slot 3 as prompted.
3. Assemble and react	Per pair: 10 uL ligand A + 10 uL ligand B + 20 uL toluene + 10 uL Pd stock . Reference: 20 uL L17 + 20 uL toluene + 10 uL Pd . Seal, move to slot 10 and run 95 C / 500 rpm / 4 h , using the validated thermal-lag procedure.
4. While reacting	At the App pause, predose 20 uL SM/IS/DCM into ten second-round wells using <code>round2_predose.csv</code> . Remove that array to the approved ventilated ambient-evaporation location; resume promptly. Run the 14 calibration injections in parallel. Cap/remove volatile sources at the permitted pauses; remove covers before any aspiration.
5. Cool	At the end of four hours, follow the manual-move prompt and thermal-handling SOP. Move the sealed block off the heater to slot 3. Cool before opening.
6. Work up	Load LC plate in slot 6 and run <code>round1/workup.py</code> . Add 50 uL workup solvent, without additional IS , then shake the reaction block 2 min without heating using the released method. Return to slot 3.
7. Dilute and analyze	Verify homogeneous extract. Robot adds 180 uL LC diluent , then 20 uL extract to the mapped LC wells with fresh sample tips. Seal and mix LC plate 2 min on the approved shaker. Run all 66 samples using <code>lc_sequence.csv</code> .
8. Trigger next batch	Export the complete integrated peak list and publish its completion marker through the configured local hook. Wait for QC to pass and for <code>next_design.csv</code> / <code>next_dosing.csv</code> (ten new pairs). Follow the configured round-two export to obtain the released local protocols and deck-load CSV. If QC or export stops, resolve it before dosing.
9. Second reaction	Check the ten predosed wells and orientation; verify DCM is gone . Load stage quantities from the new deck CSV. Run the released second-round dosing protocol: add the selected ligands, toluene and Pd as in step 3; do not redose SM/IS or perform a second heater dry-down . Seal and react at 95 C / 500 rpm / 4 h .
10. Finish	Cool sealed, then repeat steps 6-7 with the second-round workup protocol for ten samples . Stop the clock when all ten analyses are complete. Save calibrated results, QC, stock/refill records and Opentrons JSON logs.

Pause and correct before resuming

- A tip hits the bottom, touches recipient liquid during dispensing, aspirates air, drips unexpectedly, or shows bubbles/solids: pause; correct the cause. Do not force repeated aspirations through an insoluble stock. Starting heights (1 mm above bottom; dispense 2 mm above top) require actual clearance and immersion validation.
- Reuse a dosing tip only for its same stock with validated noncontact dispensing. Sample tips stay fresh between wells/columns. Replace racks when prompted; recap stocks at permitted pauses.
- If the extract is visibly two-phase or contains solids, stop sampling and use the approved extraction method. Keep the reaction sealed through heating/cooling; open only under HTE's thermal-handling SOP.

Per reaction: 5 umol SM, 50 uL final toluene solution, 0.1 M; 6 mol% each ligand, 12 mol% Pd; no base or external fluoride. **Timing:** 2 min per LC sample; 76 reaction samples total. The provisional 729-min plan must be shortened and measured before release against the 720-min limit.

Detailed procedures and unresolved settings remain in the [reference handbook](#) and [HTE configuration checklist](#).